

AIS-radar device — Wiring & Power Guide

Connecting a Wegmatt dAISy 2+ AIS receiver to a Waveshare ESP32-S3-Touch-AMOLED-1.75-G, battery-powered, three wires, no step-up converter.

The short version. The dAISy 2+ runs on **3.3 V** (internal switch) and its data pin is 3.3 V logic. The AMOLED board's 8-pin header provides a **3V3** rail and a free GPIO. So the whole connection is **three wires** from the header to the dAISy AUX header — **no 5 V, no step-up converter, no second battery, no 9 V**. One 3.7 V LiPo powers everything; the AMOLED's USB-C charges it.

1. What you need

Item	Notes
Waveshare ESP32-S3-Touch-AMOLED-1.75-G	The GPS (-G) variant, running the ais-radar firmware. Has an 8-pin header H2 on the back edge.
Wegmatt dAISy 2+ AIS receiver	Has a 6-pin AUX SERIAL header on the rear and an internal 3.3V/5V VCC switch.
3.7 V LiPo, MX1.25 2-pin plug	Plugs into the board's BAT (J1) connector. Powers the whole device.
VHF / AIS antenna	Into the dAISy SMA. Any 162 MHz-capable antenna; see §7. Needed for real reception (not for the bench test).
3 × thin wires + soldering iron	To the H2 header pins (2.54 mm pitch). Dupont jumpers work if you fit a header.

Not needed: the DC-DC step-up (MT3608) — leave it in the drawer. A separate USB cable/powerbank for the dAISy — not used. Series LiPos for 9 V — a misunderstanding; the dAISy never needs 9 V.

2. The AMOLED 8-pin header (H2)

On the back edge of the board, silkscreened left-to-right:

IO18 IO17 IO16 RXD TXD 3V3 GND VBUS

- **IO16** — free GPIO. **This is our AIS data input.** (IO17/IO18 are taken by the onboard LC76G GPS once you solder its R15/R16 jumpers, so don't use them for AIS.)
- **3V3** — the board's always-on 3.3 V rail (present on battery *and* USB). Powers the dAISy.
- **GND** — common ground.

- **VBUS** — this is the "5 V" pin, but it is **USB-only** (dead on battery). We do **not** use it.

3. The dAISy 2+ AUX SERIAL header

6-pin header on the rear, silkscreened:

NC	TXD	RXD	GND	VCC	I/O
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dAISy pin	Use
TXD	AIS data out (3.3 V logic, 38400 baud). → goes to the ESP IO16 .
GND	Ground. → ESP GND .
VCC	Power in when fed externally. → ESP 3V3 . Draws ~50 mA.
NC / RXD / I/O	Leave unconnected. (RXD would only be needed to send config to the dAISy; we configure it over USB instead.)

Set the dAISy internal VCC switch to 3.3 V before wiring. It's the small slide switch next to the AUX header marked AUX VCC 5V – 3V3. Powering the dAISy through VCC at 5 V while it's switched to 3.3 V (or vice-versa) can damage it. The TXD/RXD/I/O pins are 3.3 V regardless of this switch.

4. Wiring — three connections

#	From — AMOLED H2	To — dAISy AUX	Wire
1	GND	GND	Ground — connect first.
2	3V3	VCC	Power (dAISy switch = 3.3 V).
3	IO16	TXD	AIS data into the ESP UART RX.

AMOLED H2	dAISy 2+ AUX SERIAL
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GND	GND
3V3	VCC (switch: 3V3)
IO16	TXD
	RXD – (unused)
	I/O – (unused)
	NC – (unused)

Make and break connections with the AMOLED powered off (battery unplugged and off USB). Live wiring is how 3.3 V GPIOs get fried. Double-check polarity before first power-on.

5. Power summary

- **Source:** one 3.7 V LiPo into the board's **BAT** connector. The AXP2101 PMIC runs the board from it and steps it to the 3.3 V rail that also feeds the dAISy.
- **Charging:** plug the AMOLED's **USB-C** into any charger — the AXP2101 charges the LiPo. No separate charger, no series cells.
- **Budget:** dAISy ~50 mA at 3.3 V, comfortably within the board's 3.3 V rail (which already drives the AMOLED + GPS). The dAISy's VCC pin is rated 150 mA at 3.3 V, so there's margin.
- **Why not the header's VBUS/5 V?** It only exists while USB-C is plugged in. On battery it's 0 V, so it can't power the dAISy — hence we use the 3V3 rail instead.

If you ever did want 5 V (e.g. to run the dAISy off its own USB or a future 5 V sensor): the step-up module is fed from the battery or the 3V3 pin and trimmed to 5.0 V with a meter *before* connecting the load. Not needed for this build.

6. dAISy configuration & bench test

- **Baud:** the dAISy outputs NMEA-0183 AIVDM at **38400** baud by default — the firmware's AIS UART is set to match. No change needed.
- **Bench test without an antenna or live traffic:** connect the dAISy to a computer over USB, open its serial console, and press **T** (test mode). It then emits a test AIS message every 5 s. With the wiring above, that target should appear on the AMOLED — proving the full chain before you rely on real reception.
- **Real reception** needs the VHF antenna and line-of-sight to traffic (see §7).

7. Antenna (for real AIS)

- Any antenna tuned to the marine VHF band (~162 MHz) works; a VHF whip (~90 cm) gives far better range than a short "rubber duck". Height + line-of-sight to the water matter most.

Never connect the dAISy directly to the same antenna as a transmitting VHF radio — it will damage the receiver. Use an **active** AIS/VHF splitter if sharing.

8. Firmware side (for reference)

The ais-radar device firmware opens a hardware UART on **IO16** at **38400** baud and feeds the bytes into the salvaged AIVDM decoder (shared/ais/AisTargetDecoder), which now records target **lat/lon** as well as SOG/COG. No NMEA 2000, no WiFi — the targets are drawn radar-style on the AMOLED and (later) sent to the iOS app over BLE.

